

TrendSpot - News Aggregator App

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled “TrendSpot-News Aggregator App” is the Bonafide work of “KARTHIKA LAKSHMI V(230701142)” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The rapid growth of digital media has increased the need for quick and convenient access to news, but users often depend on multiple platforms such as websites, mobile applications, and social media to stay updated, which makes the process time-consuming and inefficient. To overcome this problem, the project titled “TrendSpot – News Aggregator App” is developed as a centralized mobile application that provides real-time news updates from various categories in one place. The application is built using Flutter, which ensures a smooth and responsive user interface, and Firebase Authentication is used to provide secure login functionality. The app integrates News API to fetch live news data, which is processed and displayed in an organized and user-friendly format. Users can explore different categories such as Sports, Technology, Movies, and Trending topics, making it easy to access information based on their interests. The application also includes features that allow users to read full news articles within the app, open the original source in a browser, and save important news using a favorites option for future reference. This favorite feature is implemented using local storage through Shared Preferences, ensuring fast access and offline availability. The user interface is designed to be simple, clean, and intuitive, allowing easy navigation and enhancing the overall user experience. The application focuses on providing real-time updates, minimal loading time, and a visually appealing layout suitable for all users.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

The project titled **“TrendSpot – News Aggregator App”** is developed to provide users with a simple and efficient way to access the latest news in one place. In today’s digital era, people rely on multiple platforms to stay updated, which can be time-consuming and inconvenient. This application addresses that problem by collecting real-time news from various sources and presenting it in a single, user-friendly interface. Users can browse news based on categories such as Sports, Technology, Movies, and Trending topics, making it easier to access information according to their interests.

The application is developed using Flutter for the frontend and Firebase Authentication for secure user login. It uses News API to fetch real-time news data and displays it in an organized format. Additional features such as reading full articles, saving news to favorites, and opening original sources enhance the user experience. The app is designed with a clean and intuitive interface, ensuring easy navigation and smooth performance. Overall, the project demonstrates how modern technologies can be integrated to build an efficient and practical mobile application for everyday use.

1.2SCOPE OF THE WORK

The scope of the project includes the design and development of a mobile news aggregator application that provides real-time news

updates from various categories such as Sports, Technology, Movies, and Trending topics. The application focuses on delivering a simple and user-friendly interface where users can easily browse news, read full articles, and stay updated with the latest information. It also includes features like category-based filtering, opening original news sources, and saving important articles for future reference using local storage.

The project is limited to mobile application development using Flutter, with Firebase used for user authentication and News API for fetching data. The scope does not include advanced features such as AI-based recommendations, cloud synchronization of saved data, or multi-language support. However, the application provides a strong foundation for future enhancements, where additional features and improvements can be integrated to expand its functionality and user experience.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The aim of this project is to develop a mobile news aggregator application called TrendSpot that provides real-time news updates in a simple and efficient manner. The main objectives are to create a user-friendly interface, integrate News API for fetching live news, implement secure user authentication using Firebase, and provide features such as category-based news browsing, reading full articles, and saving news to favorites. The project also aims to improve user convenience by offering all trending updates in one place, reducing the need to use multiple applications. In addition, the application focuses on delivering fast performance, easy navigation, and a clean user interface to enhance the overall user experience. It also ensures efficient data handling and quick access

to information, making it suitable for everyday use. Furthermore, the project serves as a foundation for future enhancements such as personalized recommendations, cloud storage, and advanced filtering options. . It focuses on delivering category-based news, enabling users to read and save articles, and improving convenience by providing all trending updates in one place.

CHAPTER 2

LITERATURE SURVEY

A mobile news aggregator application that provides real-time updates using API integration is discussed in [1]. It collects news from multiple sources and displays them in a categorized format such as sports, technology, and entertainment. The system ensures faster access to information through a responsive user interface. Data is fetched using REST APIs and displayed dynamically, improving user engagement. This study supports the News Fetching and Category Module of TrendSpot. A secure user authentication system using Firebase Authentication is presented in [2]. It allows users to sign up and log in using email and password while ensuring data privacy and security. The system is lightweight and easy to integrate with mobile applications. It provides session management and secure access control. This research supports the Authentication Module in TrendSpot. A mobile application using API-based data retrieval and JSON parsing is proposed in [3]. It processes real-time data efficiently and updates the UI dynamically based on user interactions. The system ensures smooth performance and quick response time. This approach improves data handling and enhances the user experience. This study supports the API Integration and Data Handling Module in TrendSpot. A user-friendly mobile UI

design with category-based filtering and navigation is explained in [4]. It focuses on simplicity, clean layout, and easy accessibility of information. The design ensures that users can quickly browse and read content without complexity. A study shows improved usability and user satisfaction. This research guides the UI/UX Design Module in TrendSpot. A local storage mechanism using SharedPreferences for saving user data is introduced in [5]. It allows users to store important information such as favorites and retrieve it quickly without internet access. The system ensures fast performance and efficient data management.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

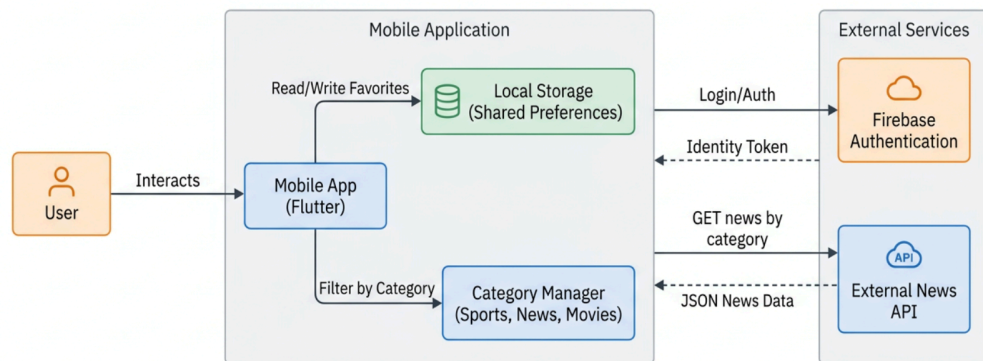


Fig 3.1.1 – Architecture Diagram

The architecture of the TrendSpot application follows a simple client-server model. The user interacts with the mobile application built using Flutter, which acts as the frontend. When the user logs in, Firebase Authentication handles secure user verification. After

successful login, the app sends requests to the News API based on the selected category. The API processes the request and returns news data in JSON format, which is then displayed in the application in an organized and user-friendly interface.

The application also includes local storage using SharedPreferences to manage saved news (favorites). When a user saves an article, the data is stored locally on the device, allowing quick access even without an internet connection. The overall architecture ensures smooth communication between the user interface, external API, and local storage, providing efficient performance, real-time updates, and a seamless user experience.

3.2 USE CASE DIAGRAM

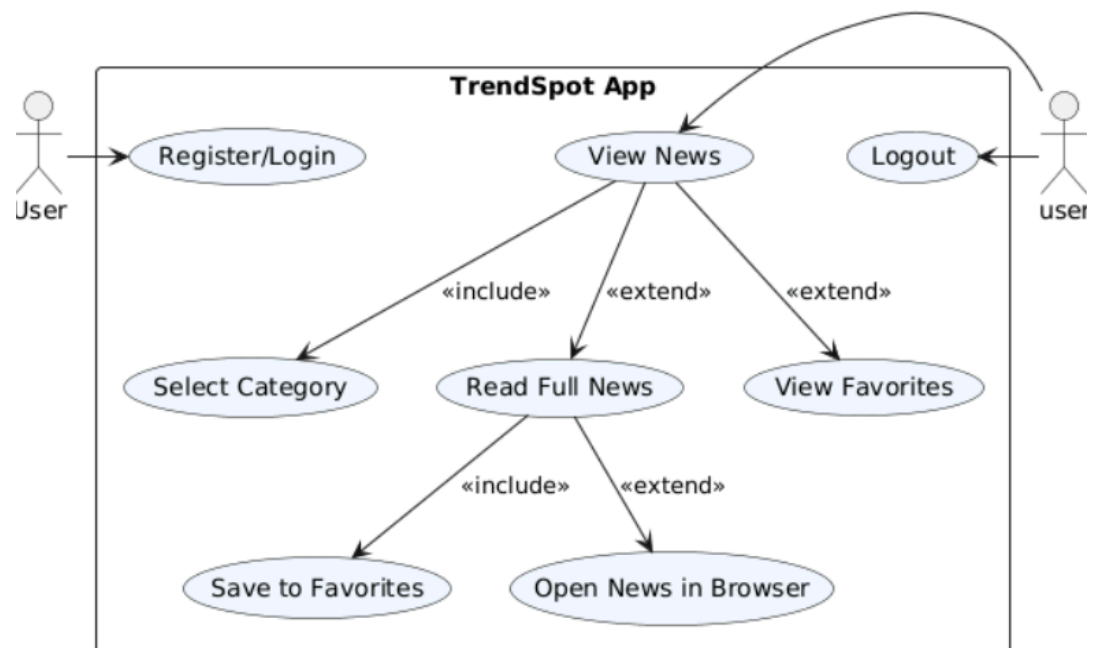


Fig 3.2.1 – Use Case Diagram

The use case diagram of the TrendSpot application represents the interaction between the user and the system. The primary actor in this system is the user, who performs actions such as registering or

logging into the application using Firebase Authentication. After successful login, the user can view news articles, select different categories like Sports, Technology, Movies, and Trending, and access real-time updates fetched from the News API. The system processes user requests and displays the relevant information in a structured format.

In addition to viewing news, the user can read full articles, save important news to favorites, and access them later. The application also allows users to open original news sources in an external browser for more details. The logout functionality ensures secure session management. Overall, the use case diagram illustrates how the user interacts with various features of the application, ensuring a smooth and user-friendly experience.

3.3 HARDWARE SPECIFICATIONS

The TrendSpot mobile application is designed to operate efficiently on standard smartphones and computing devices without requiring specialized or high-end hardware. The system architecture is optimized to support essential functionalities such as real-time news fetching, category-based filtering, user authentication, and favorites management with minimal resource consumption. The application is lightweight and ensures smooth performance even on mid-range devices. It is developed using Flutter, which provides a responsive and consistent user interface across different platforms. The integration of Firebase Authentication and News API allows secure user access and fast data retrieval without placing heavy demands on device resources. The application is designed to handle dynamic content loading and real-time updates while maintaining

performance stability. The following hardware specifications are recommended for optimal performance:

- Processor: Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher, MediaTek Helio P60 or higher)
- RAM: Minimum 3 GB (Recommended: 4 GB or higher for better multitasking and smoother performance)
- Storage: 32 GB internal storage or higher (to store application data, cached news, and user preferences)
- Graphics: Integrated graphics sufficient for rendering UI components (no dedicated GPU required)
- Operating System: Android 8.0 (Oreo) or higher for compatibility and security updates
- Display: Minimum 5-inch touchscreen display with standard resolution for better user interaction

These specifications ensure that the TrendSpot application runs efficiently across a wide range of devices, providing fast loading of news content, smooth scrolling, responsive user interactions, and reliable performance.

3.4 SOFTWARE SPECIFICATIONS

The TrendSpot mobile application is designed using a modular software architecture to ensure smooth operation across various devices. The app integrates features such as real-time news fetching, category-based filtering, user authentication, favorites management, and external news access. The software stack is composed of the following technologies:

- **Frontend Framework:**

Flutter (for building a responsive and cross-platform user interface with smooth performance)

- **Styling Tools:**

Material Design (for creating a clean, modern, and user-friendly UI)

- **Backend Framework:**

Firebase (for user authentication, secure login, and session management)

- **API Integration:**

News API (for fetching real-time news data from multiple sources)

- **Database / Storage:**

SharedPreferences (for storing user favorites and local app data efficiently)

- **Networking & Data Handling:**

HTTP package (for sending API requests and receiving JSON data)

Dart JSON decoding (for processing and displaying API data dynamically)

- **Development Environment:**

Visual Studio Code / Android Studio (for coding, testing, and debugging the application)

CHAPTER 4

PROPOSED SYSTEM

The Proposed System for the TrendSpot application is designed to provide a centralized platform for accessing real-time news from

multiple sources in a simple and efficient manner. The system focuses on user-centric features such as category-based news browsing, real-time updates, secure authentication, favorites management, and seamless navigation. It aims to reduce the need for multiple applications by delivering all trending news in one place with a clean and intuitive interface.

Key components of the proposed system include:

1. **User Authentication:** Allows users to securely register and log in using Firebase Authentication, ensuring safe access and session management.
2. **News Fetching Module:** Retrieves real-time news data from News API based on user-selected categories such as Sports, Technology, Movies, and Trending.
3. **Category Filtering:** Enables users to filter and view news according to their interests, improving accessibility and personalization.
4. **News Detail View:** Allows users to read full news articles within the application in a structured and readable format.
5. **Favorites Management:** Provides the ability to save important news articles for later access using local storage (SharedPreferences).
6. **External News Access:** Enables users to open the original news source in an external browser for complete information.
7. **User Interface Module:** Offers a clean, responsive, and user-friendly UI for smooth navigation and better user experience.
8. **Logout & Session Control:** Allows users to securely log out and manage their sessions effectively.

The system integrates these modules into a cohesive and efficient architecture, ensuring smooth data flow between the user interface, API services, and local storage. Real-time news data is fetched using HTTP requests and processed in JSON format for dynamic display within the app. The application uses SharedPreferences for storing user-specific data such as saved news, ensuring quick access even without internet connectivity. Firebase Authentication provides secure user management and enhances application reliability. The frontend is developed using Flutter, which ensures cross-platform compatibility, responsive design, and high performance across devices. Overall, the proposed system delivers a fast, reliable, and user-friendly solution for accessing and managing news content efficiently.

CHAPTER 5

MODULE DESCRIPTION

The TrendSpot application is designed using a modular architecture, where each module performs a specific function to ensure smooth operation and maintainability. The system is divided into multiple modules such as User Authentication, News Fetching, Category Filtering, News Detail View, Favorites Management, External Access, and User Interface. Each module works independently but is integrated to provide a seamless user experience.

1. User Authentication Module

This module is responsible for handling user registration and login functionalities. It uses Firebase Authentication to provide secure access to the application. Users can sign up using their email and password and log in to access the app features. The module ensures

secure session management and prevents unauthorized access. It also allows users to log out safely. This module enhances data security and ensures that user-specific data such as saved favorites is protected.

2. News Fetching Module

The News Fetching Module is responsible for retrieving real-time news data from the News API. It sends HTTP requests to the API based on user actions and receives data in JSON format. The module processes this data and extracts relevant information such as title, image, description, and source. It ensures that the latest news is displayed dynamically in the application. Error handling mechanisms are included to manage network failures or invalid responses, ensuring smooth app performance.

3. Category Filtering Module

This module allows users to filter news based on categories such as Sports, Technology, Movies, and Trending. When a user selects a category, the application sends a request to the News API with specific query parameters. The module ensures that only relevant news articles are displayed, improving usability and user satisfaction. This feature provides a personalized experience and helps users quickly access information based on their interests.

4. News Detail View Module

The News Detail View Module enables users to read full news articles within the application. When a user selects a news item, detailed information such as title, image, and description is displayed on a separate screen. This module improves readability

and provides a better user experience compared to viewing only headlines. It ensures proper formatting and structured presentation of content for easy understanding.

5. Favorites Management Module

This module allows users to save important news articles for future reference. It uses SharedPreferences for local storage, where saved articles are stored in JSON format. Users can view their saved news in a separate Favorites section. This module ensures quick data retrieval and provides offline access to saved content. It enhances user engagement by allowing personalization and easy access to preferred articles.

6. External News Access Module

The External Access Module allows users to open the original news source in a web browser. This is implemented using URL launching functionality. It ensures that users can access complete and detailed information beyond the app interface. This module increases credibility and provides flexibility to users to explore more content directly from the source.

7. User Interface Module

The User Interface Module is responsible for designing and managing the visual layout of the application. It is developed using Flutter and follows Material Design principles to ensure a clean, modern, and responsive interface. The module includes screens such as Login Page, Home Page, News List, and Favorites Page. It ensures smooth navigation, proper alignment of components, and an intuitive user experience.

8. Data Handling and Networking Module

This module manages data communication between the application and external services. It uses the HTTP package to send requests and receive responses from the News API. The received JSON data is decoded and processed efficiently. The module ensures fast data transfer, minimal loading time, and proper error handling. It plays a key role in maintaining application performance and reliability. All the modules in the TrendSpot application are integrated to work together efficiently, ensuring smooth functionality and enhanced user experience. The modular design makes the application scalable, maintainable, and easy to upgrade in the future. Each module contributes to the overall performance, making the system reliable and user-friendly.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

The proposed system, TrendSpot – News Aggregator App, offers several advantages by providing a centralized platform for accessing real-time news from multiple sources. One of the primary benefits is that it eliminates the need for users to switch between multiple applications or websites, thereby saving time and improving convenience. The application delivers up-to-date news instantly using API integration, ensuring that users always stay informed about current events. The category-based filtering feature allows users to view news according to their interests such as Sports, Technology, Movies, and Trending topics, which enhances

personalization and user satisfaction.

Another key benefit of the system is its simple and user-friendly interface, which makes it easy for users of all age groups to navigate and access information without difficulty. The integration of Firebase Authentication ensures secure login and protects user data, adding reliability to the system. The favorites feature allows users to save important articles for later reading, improving usability and engagement. This data is stored locally using SharedPreferences, enabling quick access even without an internet connection. Additionally, the application provides an option to open original news sources in a browser, giving users access to complete and detailed information.

The system is designed to be lightweight and efficient, ensuring smooth performance on devices with moderate hardware specifications. It uses optimized data handling techniques to minimize loading time and enhance responsiveness. The modular architecture of the application makes it easy to maintain and upgrade, allowing future enhancements such as AI-based recommendations, cloud synchronization, and multi-language support. Overall, the proposed system improves accessibility, efficiency, and user experience by delivering all trending news in a single, organized, and reliable platform.

CHAPTER-6

OUTPUT

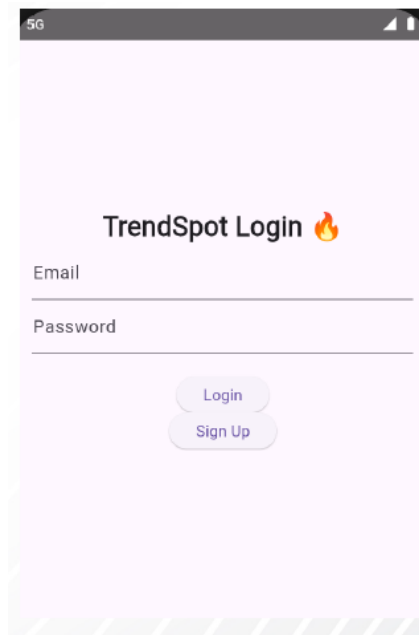


Fig 1: LOGIN PAGE

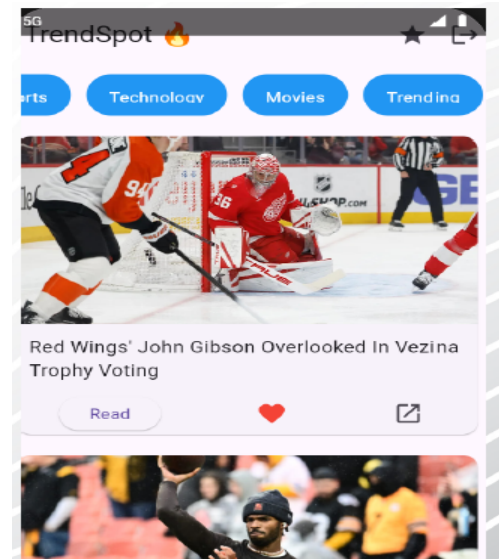


Fig 2: HOME PAGE



Fig 3:VIEW THE NEWS



Fig 4:SAVE THE FAVOURITES

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The **TrendSpot – News Aggregator App** provides users with a centralized and efficient platform to access real-time news from multiple sources in one place. With features such as category-based news browsing, real-time updates, secure user authentication, favorites management, and external news access, the app simplifies the process of staying informed. The clean and user-friendly interface ensures easy navigation, allowing users to quickly browse, read, and save news articles based on their interests. The integration of Firebase Authentication enhances security, while local storage using SharedPreferences enables offline access to saved news. The application is designed to be lightweight, fast, and responsive, ensuring smooth performance across different devices. By combining essential features into a single platform, TrendSpot improves accessibility, saves time, and enhances the overall user experience in consuming digital news.

Future Enhancements:

1. **User Personalization:** Introducing personalized news recommendations based on user interests, browsing history, and saved articles to improve user engagement.
2. **Cross-Platform Support:** Expanding the application to iOS and web platforms to provide a seamless experience across multiple devices.

3. **Cloud Synchronization:** Enabling cloud storage for saved favorites so users can access their data across devices using Firebase.
4. **AI-Based News Recommendation:** Implementing machine learning techniques to suggest relevant news articles based on user behavior and preferences.
5. **Multi-Language Support:** Adding support for multiple languages to make the app accessible to a wider audience.

CHAPTER 8

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